

NRES 746 – Workout of the Day #3: Spot the Bug

Name: _____

Instructions: Work alone for about 5 minutes on Part 1. Then get together with your neighbor(s) and spend a few minutes on Part 2.

Part 1: Find the bug

Each snippet below has one bug. None of them are typos in the strict sense – the R is syntactically valid enough to run (or at least to try to run). Circle the bug and write your fix directly on the snippet.

1a. Argument order

```
ricker <- function(a, b, x) {  
  a * x * exp(-b * x)  
}
```

```
# Evaluate the curve at x = 10, using a = 2, b = 0.5  
ricker(10, 2, 0.5)
```

1b. Plotting a curve

```
a <- 2; b <- 0.3  
curve(a * x * e^(-b * x), from = 0, to = 15)
```

1c. Evaluating across several parameter values

```
avals <- c(0.5, 1, 1.5)  
peak_vals <- sapply(avals, ricker(x = 5, b = 0.2))
```

Part 2: Write it yourself (individual, then compare with a neighbor)

2. Assume `ricker <- function(x, a, b) a * x * exp(-b * x)` is already defined. In one line of R, use `sapply()` to compute the Ricker function's value at $x = 5$ for three different values of a (0.5, 1, and 2), holding $b = 0.2$ fixed.